MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, ENVIRONMENTAL, 45°, SHIELD TERMINATION, CATEGORY 2B, NONSELF-LOCKING (MIL-C-38999 SERIES I AND II CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-85049.

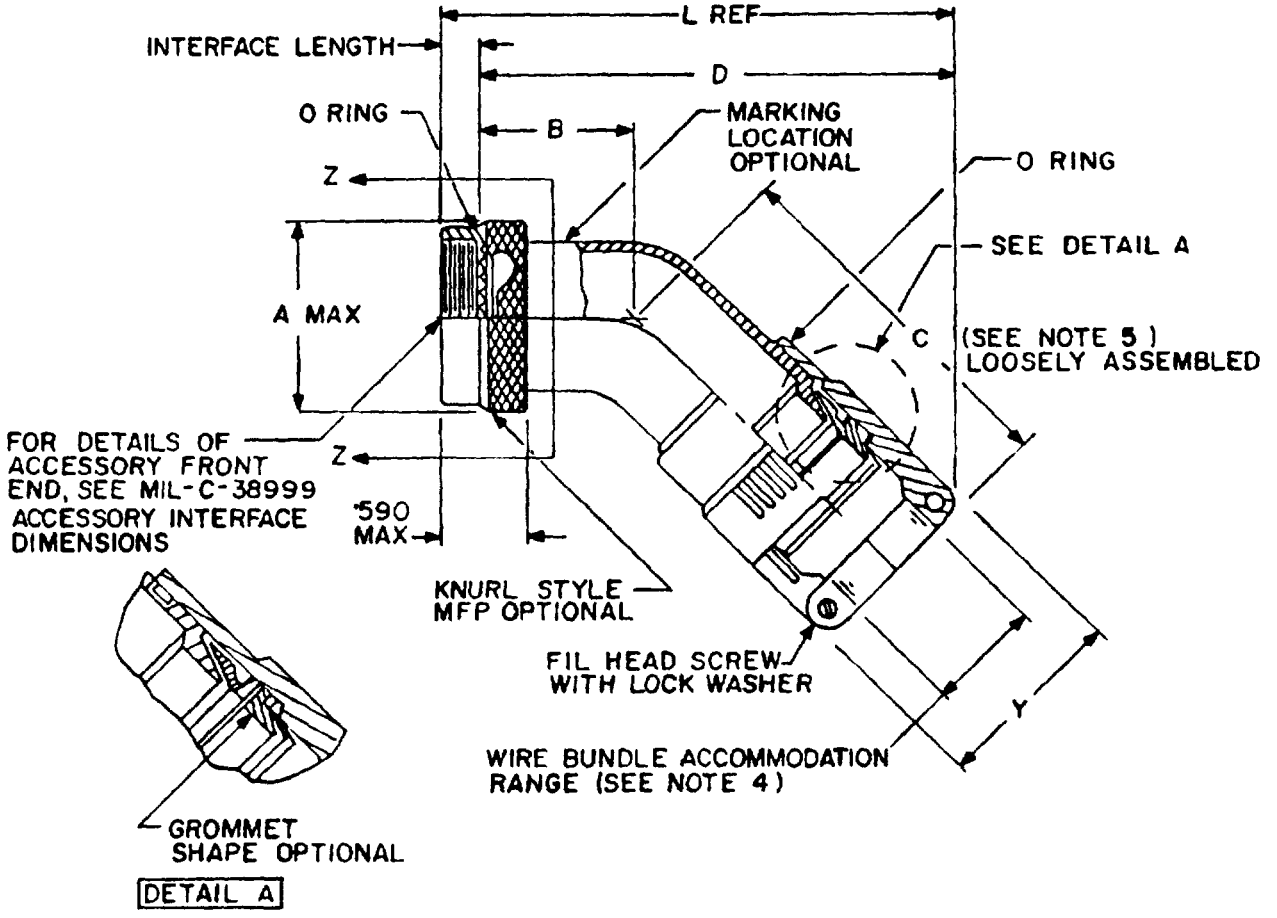
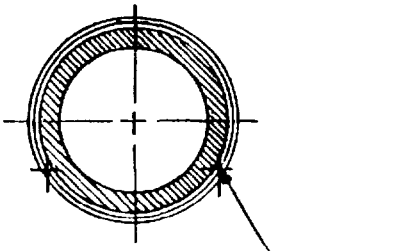


FIGURE 1. Configuration and dimensions.

DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-23T10:13Z
Check the source to verify that this is the current version before use.



3 EQUALLY SPACED
HOLES FOR .020 SAFETY WIRE

VIEW Z-Z

NOTES
1. Dimensions are in inches
2 Metric equivalents are given for general information only
3 Metric equivalents are in parentheses
4. See table I for the clamp size available with each accessory shell size. See table II for wire bundle accommodation range for each clamp size Wire bundle accommodation range is not a clamp measurement, but indicates the actual range the clamp and sealing member must accommodate. The grommet sealing dimension shall meet the maximum wire bundle accommodation range dimension of the designated clamp.
5 Loosely assembled is defined as the clamp tightened to the MIL-C-85049/77 connector accessory body two full thread lengths

FIGURE 1 Configuration and dimensions - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:13Z
Check the source to verify that this is the current version before use.

TABLE I Connector shell size and clamp size.

Accessory shell size	Fits connector shell size Series I	Fits connector shell size Series II	Clamp size (see table II) 1/	A max dia of coupling ring 2/	B 2/ ±.062	C 2/ ±.062	D 2/ ±.062	L 2/ REF
08	09	08	01 through 02	.75 (19.1)	.398 (10.1)	3.088 (78 4)	2.85 (72.39)	3.15 (80 0)
10	11	10	01 through 03	.85 (21.6)	.458 (11.6)	3 148 (80.0)	2.98 (75.69)	3 24 (82 4)
12	13	12	02 through 04	1 00 (25.4)	.518 (13 2)	3.208 (81 5)	3.13 (79.50)	3.39 (86 2)
14	15	14	02 through 05	1 15 (29.2)	.588 (14.9)	3 298 (83.8)	3.32 (84 33)	3 58 (91 0)
16	17	16	02 through 06	1 25 (31 8)	.678 (17.2)	3 408 (86.6)	3.53 (89.66)	3 79 (96 4)
18	19	18	03 through 07	1 40 (35.6)	.868 (22.0)	3 598 (91.4)	3 91 (99 31)	4 17 (106 0)
20	21	20	03 through 08	1 55 (39 4)	.868 (22 0)	3 598 (91 4)	3 96 (100.58)	4.22 (107 3)
22	23	22	03 through 09	1 65 (41 9)	.958 (24 3)	3 748 (95 2)	4.19 (106 43)	4 45 (113 1)
24	25	24	04 through 10	1 85 (47 0)	.958 (24.3)	3 748 (95.2)	4 24 (107.70)	4 50 (114 4)

1/ See note 4 of figure 1.

2/ Dimensions are in inches Metric equivalents are in parentheses and are given for general information only.

Source: https://assist.dla.mil -- Downloaded: 202640-23T10:13Z
Check the source to verify that this is the current version before use.

TABLE II. Clamp size and dimensions.

Clamp size	Wire bundle accommodation range		Y dim max
	Min 1/ 2/	Max 1/ 2/	
01	.062 (1.57)	.125 (3.18)	.80 (20.3)
02	.125 (3.18)	.250 (6.35)	1.00 (25.4)
03	.250 (6.35)	.375 (9.53)	1.10 (27.9)
04	.375 (9.53)	.500 (12.70)	1.20 (30.5)
05	.500 (12 70)	.625 (15.88)	1.25 (31.8)
06	.625 (15 88)	750 (19 05)	1 40 (35.6)
07	.750 (19.05)	875 (22.23)	1 50 (38.1)
08	.875 (22 23)	1 000 (25 40)	1 65 (41.9)
09	1 000 (25 40)	1 125 (28 58)	1.75 (44.5)
10	1 125 (28 58)	1 250 (31.75)	1 90 (48.3)

1/ Dimensions are in inches Metric equivalents are in parentheses and are given for general information only

2/ See note 4 of figure 1

Source: https://assist.dla.mil -- Downloaded: 20261O-23T10:13Z
Check the source to verify that this is the current version before use.

REQUIREMENTS.

Design and construction:

Configuration and dimensions See figure 1, and tables I and II.

Interface dimensions: In accordance with MIL-C-38999, series I and II.

Coupling nut shall be captivated to the adapter body and free to rotate.

Termination method Single shield, single cone.

Simplified assembly instructions shall be shipped with the unit package.

Material:

Adapter body and coupling nut Aluminum alloy in accordance with MIL-C-85049.

Clamp screws and lockwashers Corrosion-resisting steel, passivate

Grommet and O-rings Silicone or fluorosilicone elastomer in accordance with MIL-C-85049

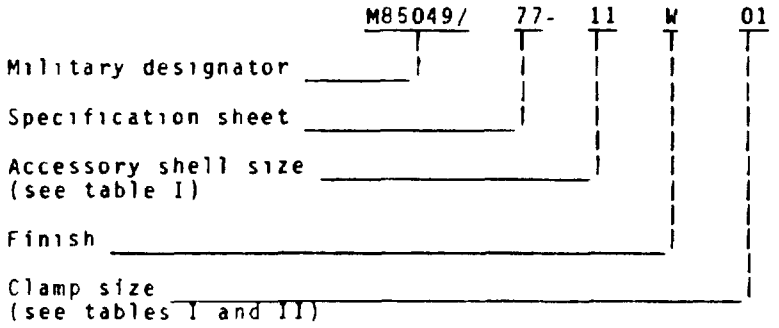
Finish In accordance with MIL-C-85049, N 1/ or W

1/ Not for Navy use Air Force use for space applications only

Source: https://assist.dla.mil -- Downloaded: 202640-23T10:13Z
Check the source to verify that this is the current version before use.

Marking In accordance with MIL-STD-1285 and as follows

Part or Identifying Number (PIN)



Qualification See MIL-C-85049, category 2B. Bending moment not applicable.

QPL evaluating activity Naval Avionics Center, Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - AS
Air Force - 85

Review activities
Army - AR, MI
Navy - EC
Air Force - 11, 15, 19, 99
DLA - ES

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5935-3636-02)

Source: https://assist.dla.mil -- Downloaded: 202610-23T10:13Z
Check the source to verify that this is the current version before use.